

World models: three paradigms

Paradigm	What is generated	Examples	Use
3D scene generation	an explicit scene (Gaussian splats / NeRF) to render	Marble (World Labs, Nov 2025), DreamFusion, GSGen	assets, simulation
interactive video generation	the next frame given the user's action	Genie 1 → 3 (DeepMind, 2024–Aug 2025), GameNGen, Oasis	playable worlds, robot data
latent world models	the next latent state, not pixels	V-JEPA 2 (Meta, Jun 2025), PAN	planning, control

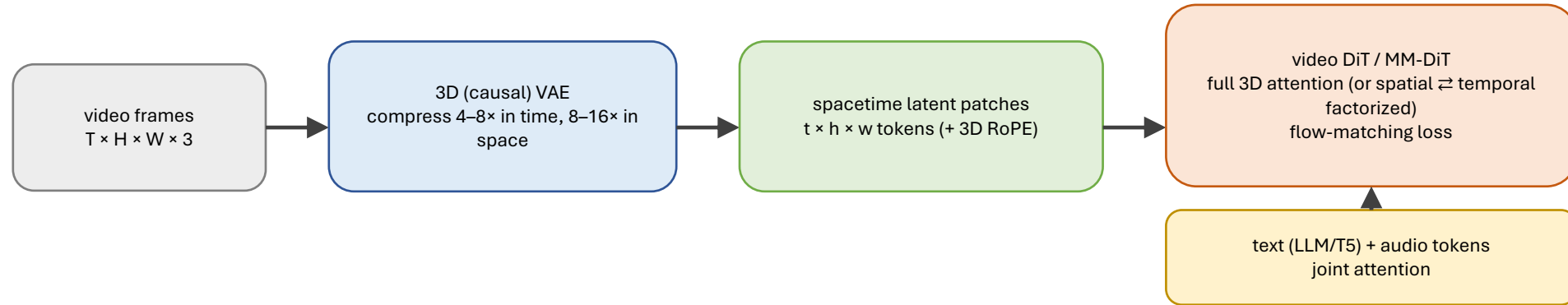
Genie 3: real-time 720p interactive worlds lasting minutes, from a text prompt; limitations reported: short memory, latent actions, no explicit geometry

A world model predicts what happens next given an action: generated 3D scenes, interactive video, or latent states

Genie (DeepMind): learns latent actions from unlabeled video; Genie 3 (Aug 2025) generates playable 720p worlds in real time

V-JEPA 2 (Meta, Jun 2025): non-generative — predicts masked latents, then plans with the predictor (ties to Lecture 5's JEPA)

Video generators are latent flow-matching transformers with a time axis

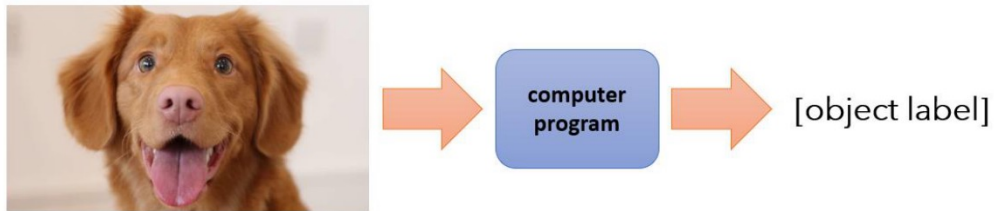


Sora (Feb 2024), Movie Gen (Oct 2024, 30B), Wan 2.x, LTX-2, Veo 3 and Sora 2 (2025) all follow this template; older: 3D U-Nets with factorized attention

A 3D VAE compresses time and space; spacetime patches become tokens with 3D positional encoding
Full 3D attention (or spatial ⇔ temporal factorized attention in older 3D U-Nets); text and now audio tokens attend jointly with video tokens
Sampling and RL tricks carry over from images: few-step distillation, CFG, reward fine-tuning

What is reinforcement learning?

supervised learning



input: $\mathbf{x}$

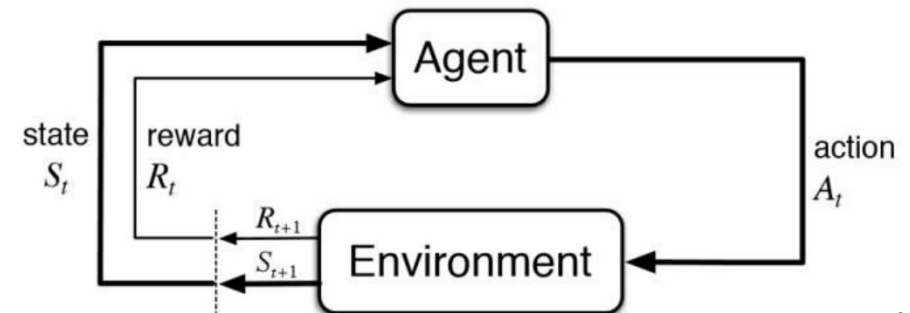
output: $\mathbf{y}$

data: $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}$

goal: $f_{\theta}(\mathbf{x}_i) \approx \mathbf{y}_i$

← someone gives this to you

reinforcement learning



input: $\mathbf{s}_t$ at each time step

output: $\mathbf{a}_t$ at each time step

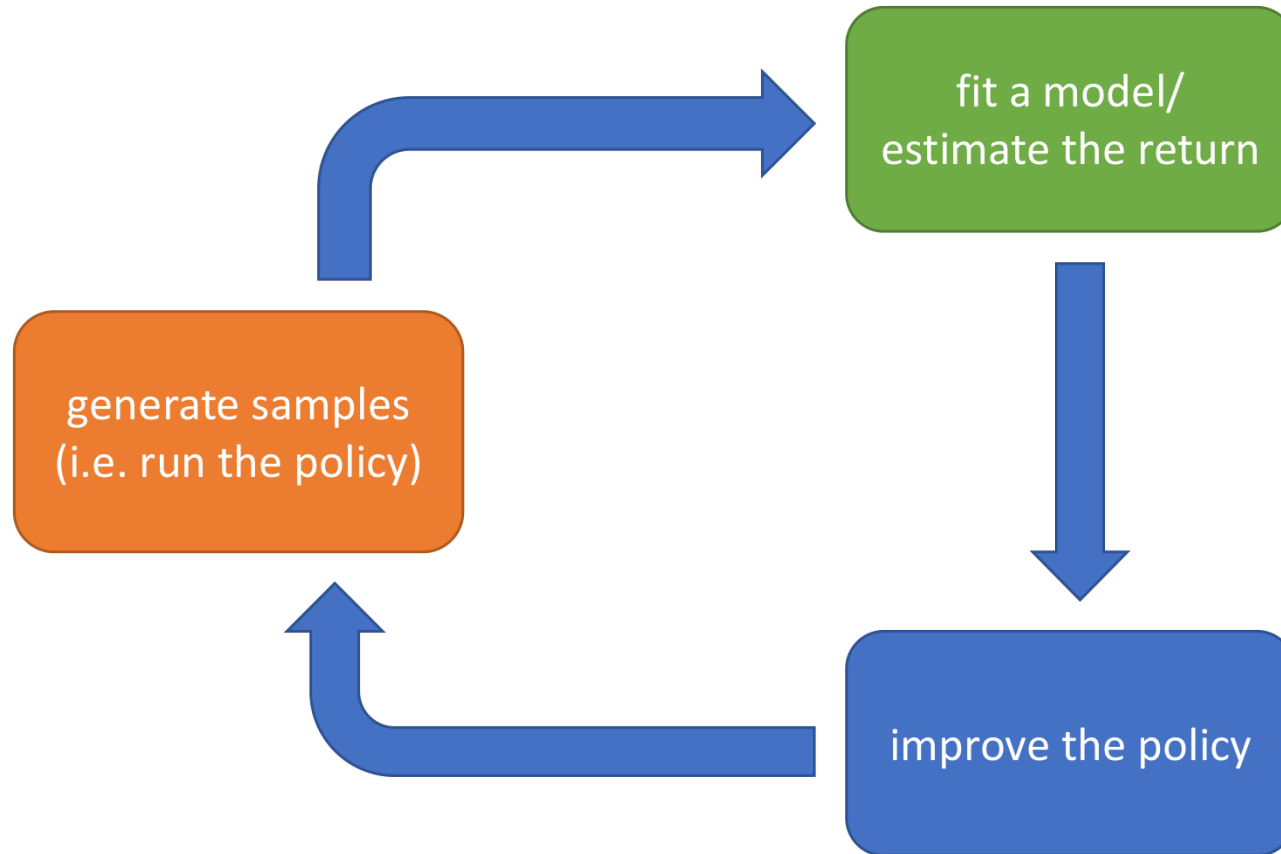
data: $(\mathbf{s}_1, \mathbf{a}_1, r_1, \dots, \mathbf{s}_T, \mathbf{a}_T, r_T)$

goal: learn $\pi_{\theta} : \mathbf{s}_t \rightarrow \mathbf{a}_t$

to maximize $\sum_t r_t$

pick your own actions

The anatomy of a reinforcement learning algorithm

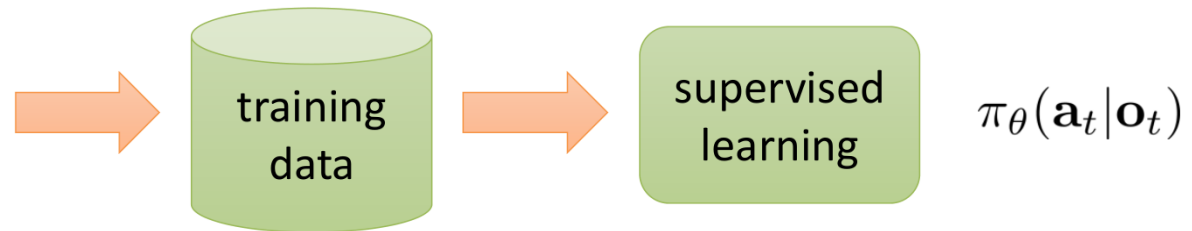
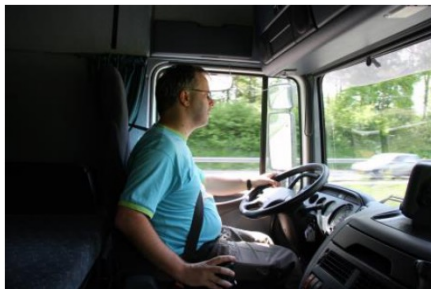
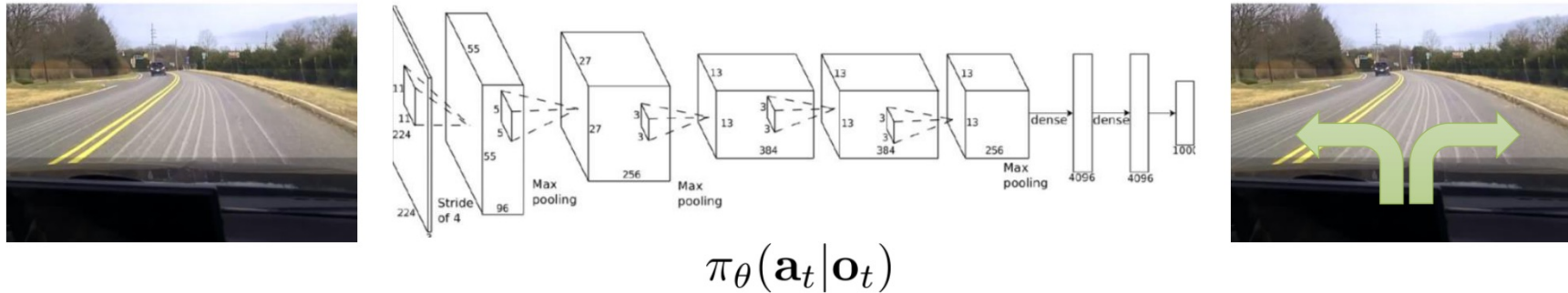


Types of RL algorithms

$$\theta^* = \arg \max_{\theta} E_{\tau \sim p_{\theta}(\tau)} \left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

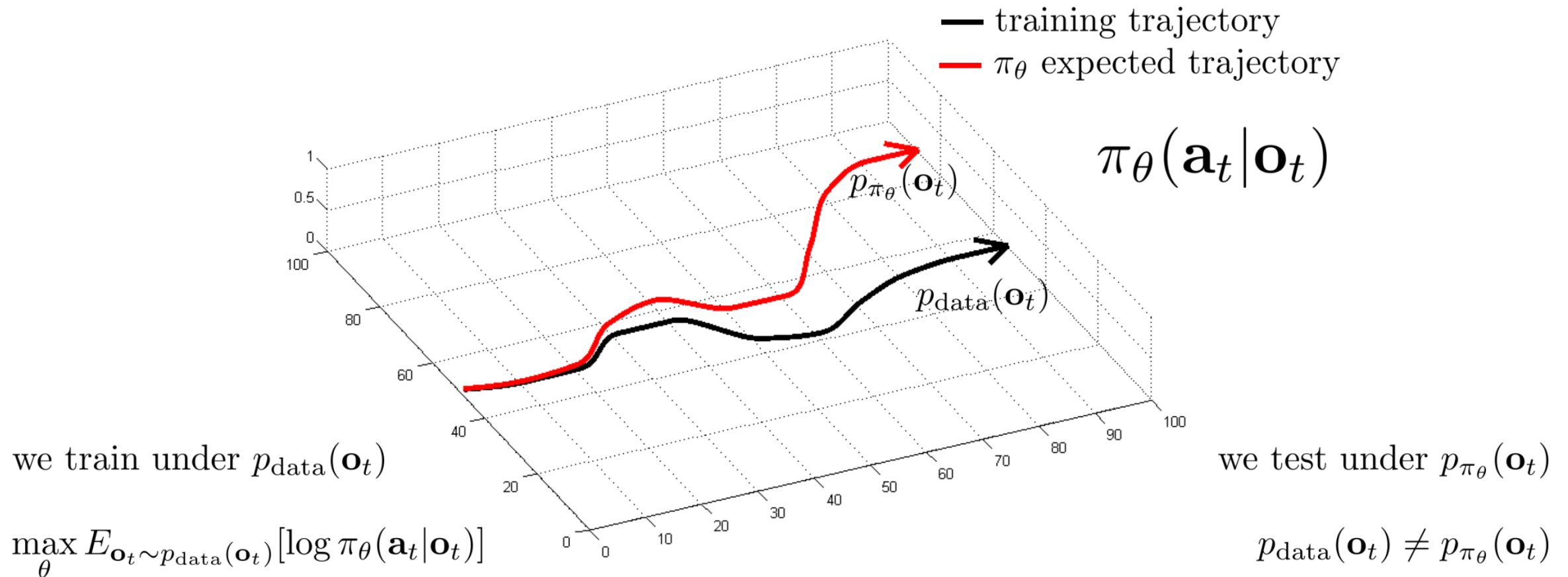
- Policy gradients: directly differentiate the above objective
- Value-based: estimate value function or Q-function of the optimal policy (no explicit policy)
- Actor-critic: estimate value function or Q-function of the current policy, use it to improve policy
- Model-based RL: estimate the transition model, and then...
 - Use it for planning (no explicit policy)
 - Use it to improve a policy
 - Something else

Imitation Learning



behavioral cloning

The distributional shift problem

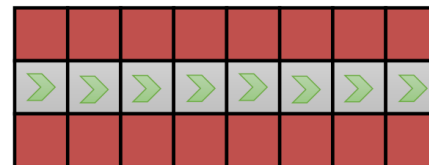
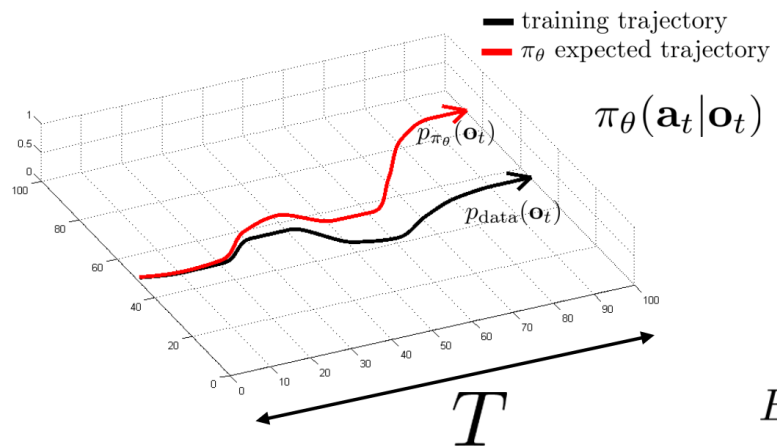


Some analysis

$$c(\mathbf{s}, \mathbf{a}) = \begin{cases} 0 & \text{if } \mathbf{a} = \pi^*(\mathbf{s}) \\ 1 & \text{otherwise} \end{cases}$$

assume: $\pi_\theta(\mathbf{a} \neq \pi^*(\mathbf{s}) | \mathbf{s}) \leq \epsilon$

for all $\mathbf{s} \in \mathcal{D}_{\text{train}}$



$$E \left[\sum_t c(\mathbf{s}_t, \mathbf{a}_t) \right] \leq \underbrace{\epsilon T}_{O(\epsilon T^2)} + \underbrace{}_{T \text{ terms, each } O(\epsilon T)}$$



Can we make it work more often?

can we make $p_{\text{data}}(\mathbf{o}_t) = p_{\pi_\theta}(\mathbf{o}_t)$?

idea: instead of being clever about $p_{\pi_\theta}(\mathbf{o}_t)$, be clever about $p_{\text{data}}(\mathbf{o}_t)$!

DAgger: Dataset Aggregation

goal: collect training data from $p_{\pi_\theta}(\mathbf{o}_t)$ instead of $p_{\text{data}}(\mathbf{o}_t)$

how? just run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$

but need labels $\mathbf{a}_t$!

- 
1. train $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$
 2. run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ to get dataset $\mathcal{D}_\pi = \{\mathbf{o}_1, \dots, \mathbf{o}_M\}$
 3. Ask human to label $\mathcal{D}_\pi$ with actions $\mathbf{a}_t$
 4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_\pi$

Robots and LLMs learn the same way in 2025–26

- **Vision-language-action models** = behavior cloning with a VLM backbone: RT-2 (Google DeepMind, 2023) emits actions as text tokens; **$\pi 0$** (Physical Intelligence, Oct 2024) adds a flow-matching action head (Lecture 10b); **$\pi 0.5$** (Apr 2025) targets open-world generalization; **Gemini Robotics** (Mar 2025)
- Then the same second stage as LLMs: RL fine-tuning on outcomes (success detectors) or human corrections (DAgger-style)
- LLMs: SFT (imitation) $\rightarrow$ RLHF (learned reward) $\rightarrow$ RLVR (verifier reward, Lecture 12); image/video generators: DDPO / Flow-GRPO (Lecture 10b)
- Levine's summary (Lecture 1): data teaches what the world looks like; optimization is what solves new problems